

EXAM 7 – SPRING 2014

5. (3.25 points)

An insurance company has 1,000 exposures uniformly distributed throughout the accident year. The a priori ultimate loss is \$800 per exposure unit.

The expected loss payment pattern is approximated by the following loglogistic function where G is the cumulative proportion of ultimate losses paid and x represents the average age of reported losses in months.

- $G(x) = \frac{x^\omega}{x^\omega + \theta^\omega}$
- $\omega = 2.5$
- $\theta = 24$

a. (1 point)

Calculate the expected losses paid in the first 36 months after the beginning of the accident year.

b. (0.75 point)

Assume the actual cumulative paid losses at 36 months after the beginning of the accident year are \$650,000. Estimate the ultimate loss for the accident year using assumptions based upon the Cape Cod method.

c. (0.5 point)

Estimate the ultimate loss for the accident year based on the loglogistic payment model and the actual payments through 36 months, disregarding the a priori expectation.

d. (1 point)

Calculate a reserve estimate for the accident year by credibility-weighting the two estimates of ultimate loss in parts b. and c. above using the Benktander method.

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SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 5	
TOTAL POINT VALUE: 3.25	LEARNING OBJECTIVE: A1: Calculate unpaid claim estimates using credibility models. A2: Estimate parameters and unpaid claims using claims development models related to loss reserving methods such as: • Chain ladder • Cape Cod • Chain ladder plus calendar-year effects • Bornhuetter-Ferguson
SAMPLE ANSWERS	
Part a: 1 point	
<i>Sample Answer 1</i> The average age of AY losses at 36 months = $36 - 6 = 30$ $G(x) = x^w / [x^w + \theta^w] = 30^{2.5} / [30^{2.5} + 24^{2.5}] = 0.636$ We would expect $1,000 \times \\$800 \times 0.636 = \\$508,800$ to be paid by 36 months <i>Sample Answer 2</i> Time = 36 Months Average Age = 30 months $G(x) = 30^{2.5} / [30^{2.5} + 24^{2.5}] = 0.636$ Expected losses after 36 months = $\\$800 \times 0.636 \times 1,000 = \\$508,800$	
Part b: 0.75 point	
<i>Sample Answer 1</i> Ultimate = Paid + Expected Unpaid Ultimate = $650,000 + (1 - 0.636) \times 800,000 = \\$941,200$ <i>Sample Answer 2 (using truncation)</i> Test for Truncation --> $g(66) = .926$ --> Need to truncate at twice 3 years = 6 years Ultimate = $650,000 + (.926 - .636) \times 800,000 = \\$882,000$	
Part c: 0.5 point	
<i>Sample Answer 1</i> Ultimate = $650,000 / 0.636 = \\$1,022,013$	

SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample Answer 2 (using truncation)

$$\text{Ultimate} = 650,000 \times (.926 / .636) = \$946,384$$

Part d: 1 point

Sample Answer 1

The credibility weight (z) for the Benktander method is $G(30) = 0.636$

$$U_{GB} = U_{LDF} \times z + (1 - z) \times U_{CC}$$

$$U_{GB} = 1,022,013 \times 0.636 + (1 - 0.636) \times 941,200 = 992,597$$

$$R_{GB} = 992,597 - 650,000 = \$342,597$$

Sample Answer 2

$$R_{GB} = R_{LDF} \times z + (1 - z) \times R_{CC}$$

$$R_{GB} = (1,022,013 - 650,000) \times 0.636 + (1 - 0.636) \times (941,200 - 650,000) = \$342,597$$

Sample Answer 3

$$R_{GB} = (1 - z) \times U_{CC} = (1 - .636) \times (941,200) = \$342,597$$

Sample Answer 4 (using truncation)

$$R_{GB} = (1 - z) \times U_{CC} = (.926 - .636) \times (882,000) = \$255,780$$

EXAMINER'S REPORT

In general, candidates scored well on this question.

Part a

The candidate was expected to know the average age of the losses, the initial ultimate loss estimate, and the application of the two to calculate estimated paid losses.

The most common mistake was candidates inappropriately calculating the average age.

Part b

The candidate was expected to know how to calculate the estimated unpaid loss, and add the actual paid losses to this amount to arrive at the ultimate.

The most common mistake was candidates using the wrong initial Ultimate loss.

SAMPLE ANSWERS AND EXAMINER'S REPORT

Part c
The candidate was expected to know to divide the actual paid loss by the estimated percent paid to date, which ignores the a-priori loss estimate.
Part d
The candidate was expected to appropriately weight the ultimate loss estimates from parts a. and b., per the Benktander method. The most common mistake made on this part of the question was not subtracting paid loss from ultimate loss in order to determine the reserve estimate. Some candidates used a different method that produced the correct answer, but did not utilize the answers in parts b and c, as the question specified. Credit was not awarded for these responses.